WORKED SOLUTION

STEP II 2016, Question 2

Pure • Factor Theorem

> **STEP II 2016, Question 2**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/16-s2-q2

The key idea

Think of $(*)$ as a polynomial in c whose coefficients happen to involve a and b. The factor theorem says $c-(a+b)$ is a factor exactly when substituting $c=a+b$ gives zero. Once one factor is found, the symmetry of $(*)$ in a, b, c hands us two more for free, and a degree count finishes the factorisation. The two equations are then disguised special cases: the job is to choose a, b, c (and later d, e) to match them.

Solution

The factor $a+b-c$

Write $F(a,b,c)$ for $(*)$. Regarded as a polynomial in c, F has the factor $c-(a+b)$, and hence $a+b-c$, if and only if $F(a,b,a+b)=0$. Put $p=a+b$ and $c=p$. Then $a+b+c=2p$, and using $a^2+b^2=p^2-2ab$ and $a^3+b^3=p^3-3abp$,

$$a^2+b^2+c^2=2p^2-2ab, \qquad a^3+b^3+c^3=2p^3-3abp.$$

So

$$\begin{aligned} F(a,b,a+b) &= 8p^3-6(2p)(2p^2-2ab)+8(2p^3-3abp) \\ &= 8p^3-24p^3+24abp+16p^3-24abp=0, \end{aligned}$$

and $a+b-c$ is a factor of $(*)$, as required.

Factorising $(*)$ completely

F is unchanged by any rearrangement of a, b, c. Swapping b and c turns the factor $a+b-c$ into $a+c-b$; swapping a and c turns it into $b+c-a$. So all three of

$$a+b-c, \qquad b+c-a, \qquad c+a-b$$

are factors. They are distinct linear polynomials, none a constant multiple of another, so their product divides F. Every term of F has degree 3, so the remaining factor has

degree 0: it is a constant k, and

$$F(a,b,c) = k(a+b-c)(b+c-a)(c+a-b).$$

To find k, put $a = b = c = 1$: the left-hand side is $27 - 6\cdot 3\cdot 3 + 8\cdot 3 = -3$ and the product is 1. Hence $k = -3$ and

$$\begin{aligned}(a+b+c)^3 - 6(a+b+c)(a^2+b^2+c^2) + 8(a^3+b^3+c^3)\\ = -3(a+b-c)(b+c-a)(c+a-b).\end{aligned}$$

Part (i)

Divide the middle and last terms of the equation into the shape of $(*)$:

$$(x+1)^3 - 6(x+1)\left(x^2+\tfrac{5}{2}\right) + 8\left(x^3+\tfrac{13}{4}\right) = 0.$$

Ask yourself Which values of a, b, c make this exactly $(*)$?

Try $a = x$ with b, c constants. We need

$$b+c = 1, \qquad b^2+c^2 = \tfrac{5}{2}, \qquad b^3+c^3 = \tfrac{13}{4}.$$

The first two give $2bc = (b+c)^2 - (b^2+c^2) = -\tfrac{3}{2}$, so $bc = -\tfrac{3}{4}$. Then

$$b^3+c^3 = (b+c)^3 - 3bc(b+c) = 1 + \tfrac{9}{4} = \tfrac{13}{4},$$

so the third condition holds automatically. Now b, c are the roots of $t^2 - t - \tfrac{3}{4} = 0$, that is $(2t-3)(2t+1) = 0$, so $b = \tfrac{3}{2}$, $c = -\tfrac{1}{2}$.

With $a = x$, $b = \tfrac{3}{2}$, $c = -\tfrac{1}{2}$ the factorisation gives

$$a+b-c = x+2, \qquad b+c-a = 1-x, \qquad c+a-b = x-2,$$

so the equation is $-3(x+2)(1-x)(x-2) = 0$, and

$$x = -2, \quad x = 1, \quad x = 2.$$

Part (ii)

Write $G(a,b,d,e)$ for the four-variable expression. (Setting $c = d+e$ makes $a+b-d-e$ look like $a+b-c$, which suggests the factor; but it's just as quick to check it directly.) By the factor theorem, regarding G as a polynomial in e, it is enough to show $G = 0$ when $a+b = d+e$. Let $p = a+b = d+e$. Exactly as before, $a^2+b^2 = p^2 - 2ab$, $d^2+e^2 = p^2 - 2de$, $a^3+b^3 = p^3 - 3abp$ and $d^3+e^3 = p^3 - 3dep$, so with $q = ab+de$,

$$\begin{aligned}G &= (2p)^3 - 6(2p)(2p^2-2q) + 8(2p^3-3pq)\\ &= 8p^3 - 24p^3 + 24pq + 16p^3 - 24pq = 0.\end{aligned}$$

So $a+b-d-e$ is a factor of G.

Now G is unchanged by any rearrangement of a, b, d, e. The factor $a+b-d-e$ comes from splitting the four letters into the pairs $\{a,b\}$ and $\{d,e\}$; the other two ways of splitting them into pairs give the factors

$$a+d-b-e \quad (\text{swap } b,d), \qquad a+e-b-d \quad (\text{swap } b,e).$$

These three linear factors are distinct and none is a multiple of another, and G is homogeneous of degree 3, so $G = k(a+b-d-e)(a+d-b-e)(a+e-b-d)$ for a constant k. Putting $a=1$, $b=d=e=0$ gives $1-6+8=3$ on the left and k on the right, so

$$G = 3(a+b-d-e)(a+d-b-e)(a+e-b-d).$$

For the equation, take $a=x$ and constants b, d, e with

$$b+d+e=6, \qquad b^2+d^2+e^2=14, \qquad b^3+d^3+e^3=36.$$

The numbers 1, 2, 3 work: $1+2+3=6$, $1+4+9=14$, $1+8+27=36$. So with $a=x$, $b=1$, $d=2$, $e=3$ the equation is exactly $G=0$, which factorises as

$$3(x+1-2-3)(x+2-1-3)(x+3-1-2) = 3(x-4)(x-2)x = 0.$$

Hence

$$x=0, \quad x=2, \quad x=4.$$

Checks

- Put $a=b=1$, $c=2$ into $(*)$: $64-6\cdot4\cdot6+8\cdot10=0$, as the factor $a+b-c$ predicts.
- In (i), $x=1$ gives $8-3\cdot2\cdot7+2\cdot17=8-42+34=0$.
- In (ii), $x=0$ gives $216-6\cdot6\cdot14+8\cdot36=216-504+288=0$, and $x=4$ gives $1000-6\cdot10\cdot30+8\cdot100=0$.
- The factorisations were also confirmed by computer at a few thousand random rational values of the variables.

Common slips

- Forgetting the constant: $(*)$ is -3 times the product of the factors, not the product itself. Always fix k with a convenient substitution.
- In (ii), listing more than three factors. Swapping a and d, say, gives $d+b-a-e$, which is just $-(a+e-b-d)$; the three genuinely different factors come from the three ways of pairing up four letters.
- Expanding everything by brute force. It works, but it is long and error-prone; matching the equation to the identity is the point of the question.

Beyond STEP

The factor theorem works over any field, and here we used it for polynomials in several variables by treating all but one as constants. Symmetry plus unique factorisation of polynomials is a powerful combination: the theory of symmetric polynomials says that every symmetric polynomial can be written in terms of the elementary ones $a+b+c$, $ab+bc+ca$, abc, which is the structure hiding behind the conditions on b, c in (i). Incidentally, the product $(a+b-c)(b+c-a)(c+a-b)$ is exactly what appears in Heron's formula for the area of a triangle. For more on polynomial rings and factorisation, see Allenby, *Rings, Fields and Groups*.

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